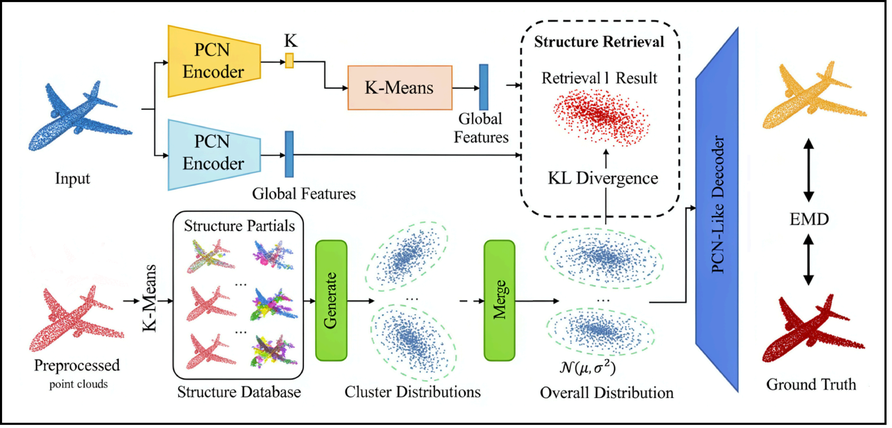
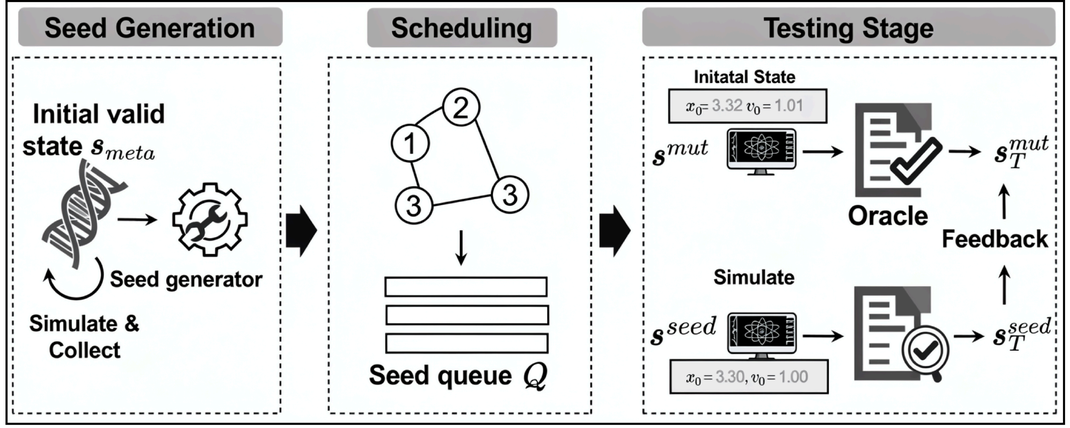
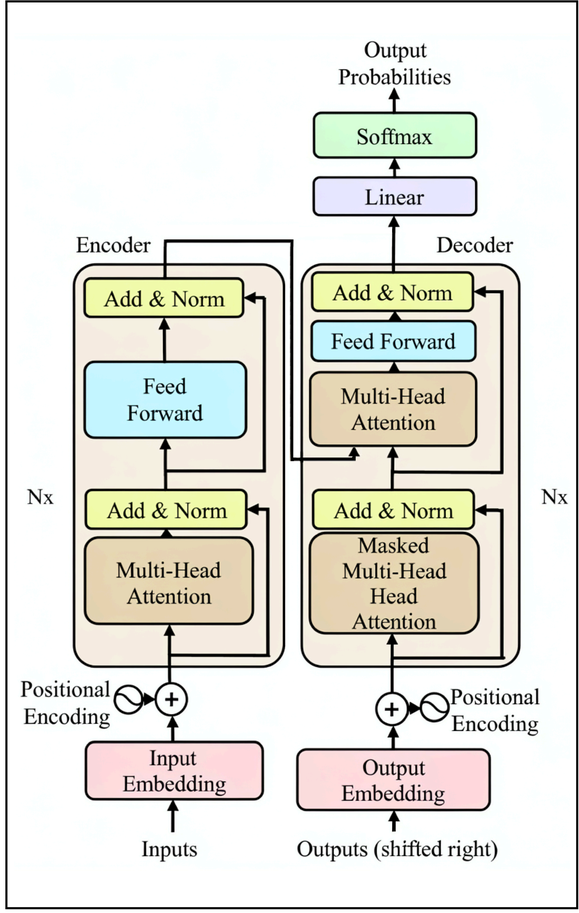
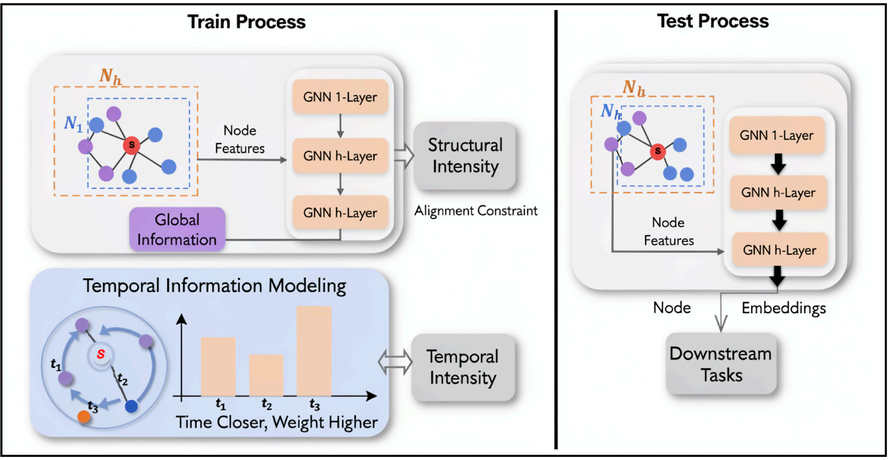
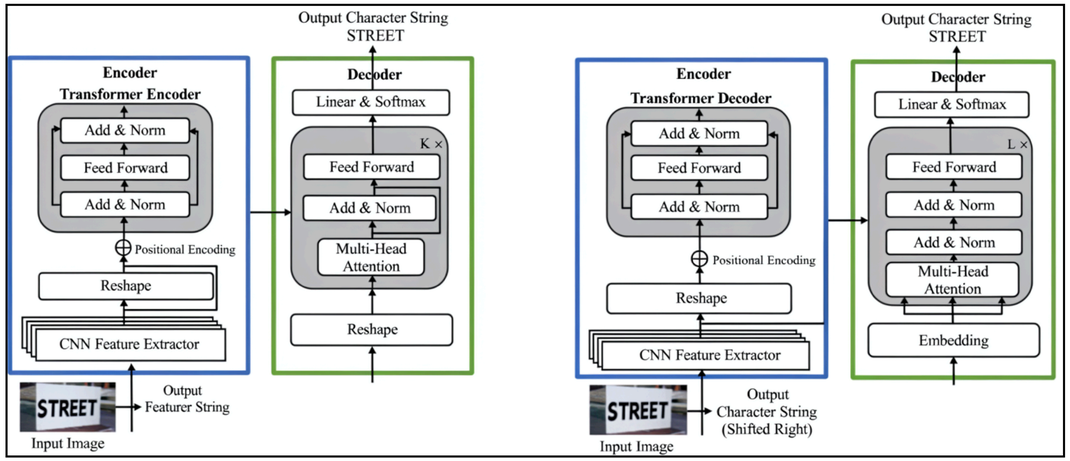
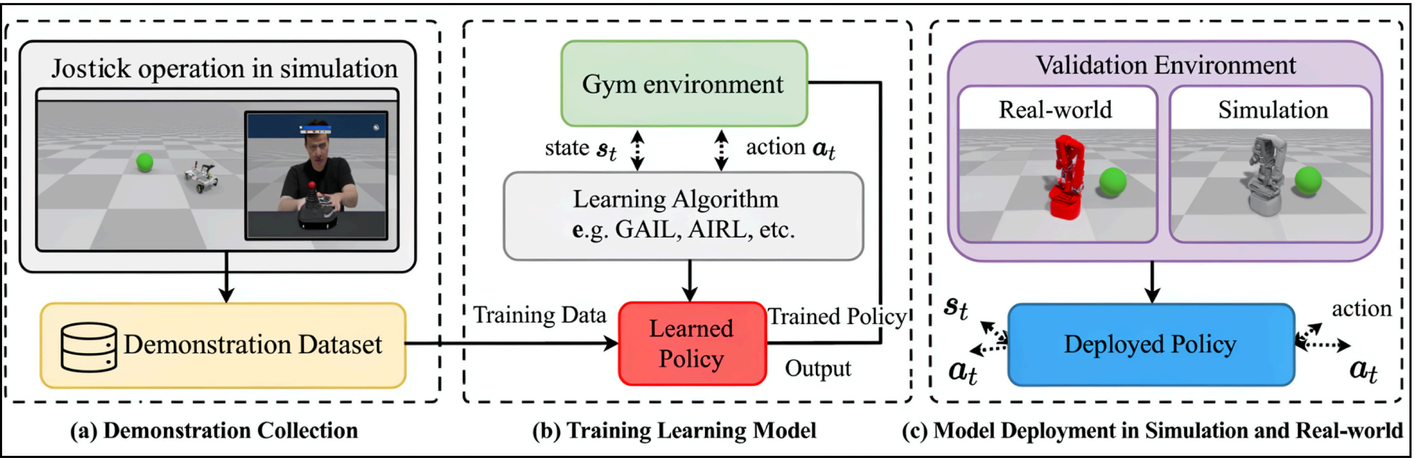

PCN Encoder
K
K-Means
Global Features
Structure Retrieval
Retrieval 1 Result
KL Divergence
PCN Encoder
Global Features
Input
PCN-Like Decoder
EMD
Ground Truth
Structure Partials
K-Means
Generate
Merge
Preprocessed point clouds
Structure Database
Cluster Distributions
$\mathcal{N}(\mu, \sigma^2)$
Overall Distribution

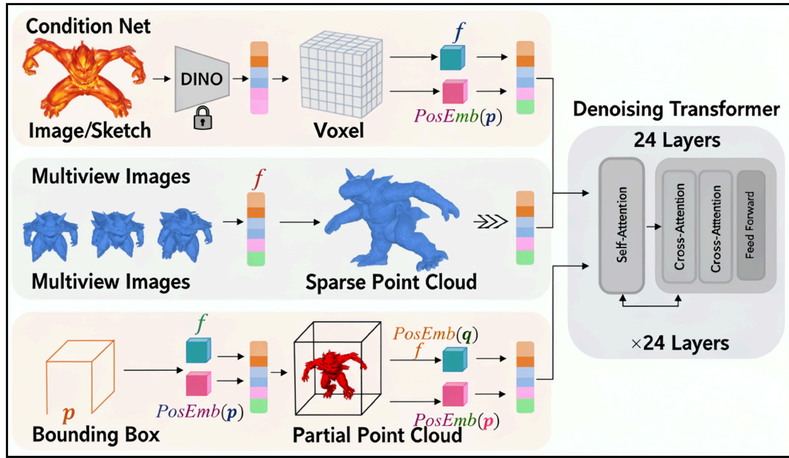

Seed Generation
Scheduling
Testing Stage
Initial valid state s_{meta}
Seed generator
Simulate & Collect
Seed queue Q
Initatal State
$x_0 = 3.32\ v_0 = 1.01$
s^{mut}
Oracle
s_T^{mut}
Feedback
Simulate
s^{seed}
$x_0 = 3.30, v_0 = 1.00$
s_T^{seed}


Output Probabilities
Softmax
Linear
Encoder
Decoder
Add & Norm
Feed Forward
Multi-Head Attention
Masked Multi-Head Head Attention
Nx
Positional Encoding
Input Embedding
Output Embedding
Inputs
Outputs (shifted right)


Train Process
Test Process
N_h
N_1
Node Features
GNN 1-Layer
GNN h-Layer
GNN h-Layer
Structural Intensity
Alignment Constraint
Global Information
Temporal Information Modeling
Time Closer, Weight Higher
Temporal Intensity
Node Embeddings
Downstream Tasks


Output Character String
STREET
Encoder
Transformer Encoder
Add & Norm
Feed Forward
Add & Norm
Positional Encoding
Reshape
CNN Feature Extractor
Decoder
Linear & Softmax
K ×
Feed Forward
Add & Norm
Multi-Head Attention
Reshape
STREET
Input Image
Output Feature String
Transformer Decoder
L ×
Embedding
Output Character String (Shifted Right)


Jostick operation in simulation
Demonstration Dataset
Gym environment
state s_t
action a_t
Learning Algorithm
e.g. GAIL, AIRL, etc.
Training Data
Learned Policy
Trained Policy
Output
Validation Environment
Real-world
Simulation
s_t
Deployed Policy
action
a_t
(a) Demonstration Collection
(b) Training Learning Model
(c) Model Deployment in Simulation and Real-world


Condition Net
DINO
Image/Sketch
Voxel
f
$PosEmb(p)$
Multiview Images
Multiview Images
Sparse Point Cloud
p
Bounding Box
$PosEmb(q)$
Partial Point Cloud
Denoising Transformer
24 Layers
Self-Attention
Cross-Attention
Cross-Attention
Feed Forward
×24 Layers